

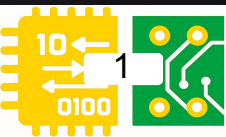
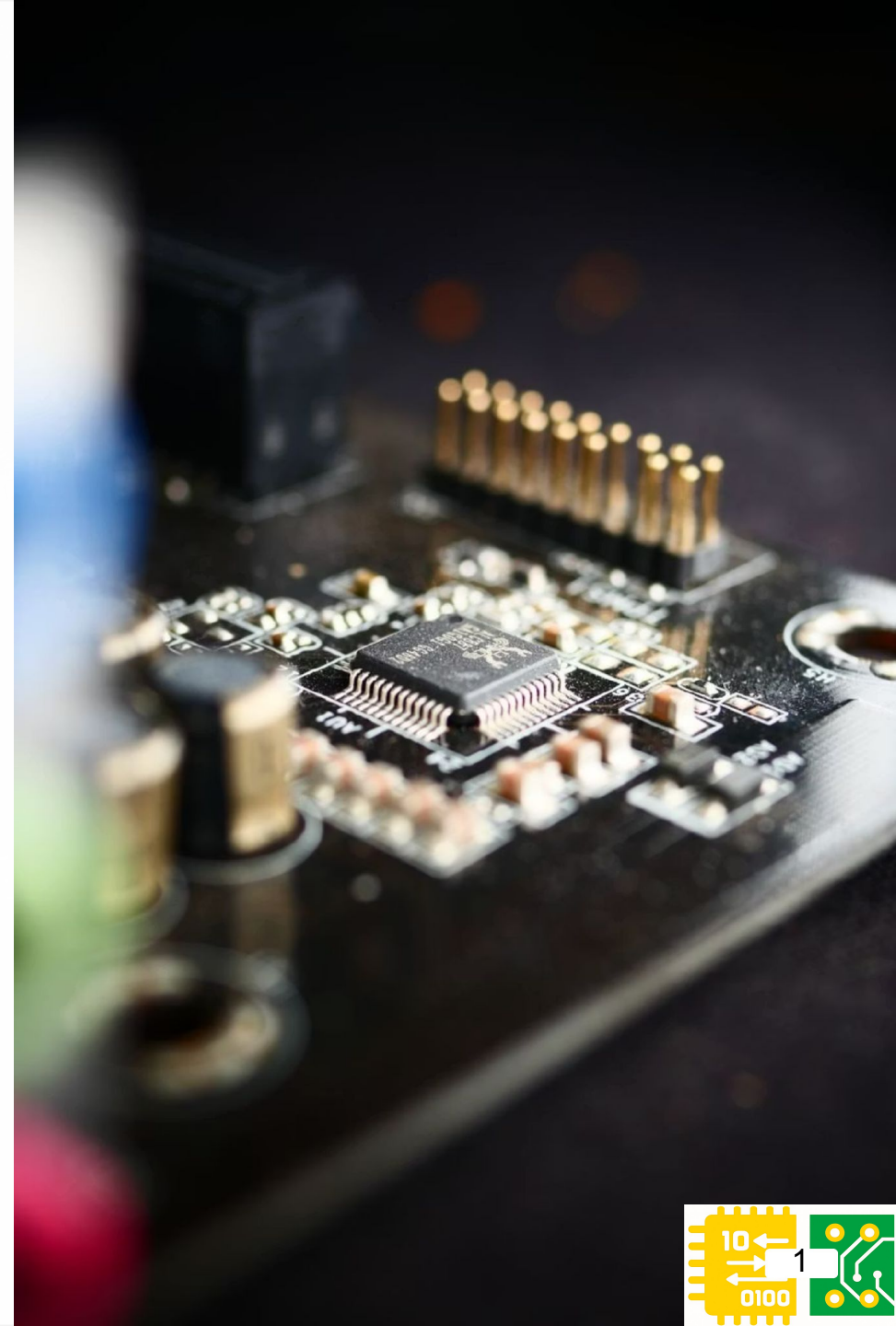
3-Floor Elevator Controller

Presented by Team A6
for Hardware Engineering Lab (SS 2026)
Prof. Dr.-Ing. Ali Hayek

PHASE 1: NEXYS A7-100T

PHASE 2: LATTICE MACHXO2 PCB

30.06.2026





Abstract & Objectives

A 3-floor elevator controller in VHDL, verified by self-checking simulation and demonstrated on a Digilent Nexys A7-100T FPGA. Control core: a three-state Mealy FSM with request latching, nearest-first scheduler, and timer-modeled motion. All seven test scenarios pass in simulation and hardware.

Floors

U (basement), E (ground), 1 (first) — one call button each with persistent request latching

Motion

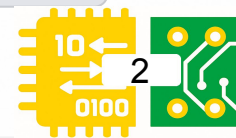
Timer-modeled travel (one floor at a time) and door dwell — no physical motor

Outputs

7-segment floor display, travel-direction LEDs, door-open LED

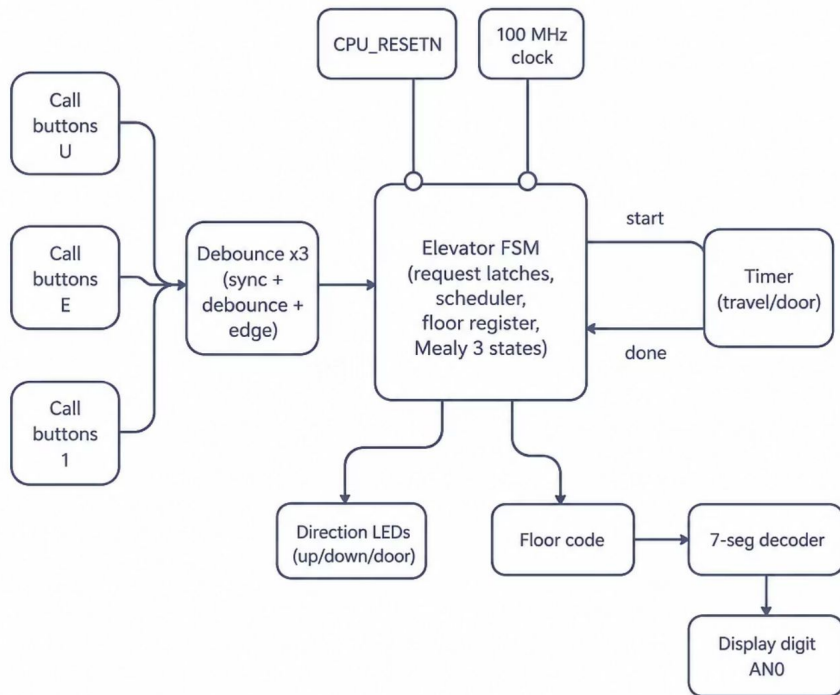
Inputs

Robust handling of asynchronous, bouncy button inputs via debounce



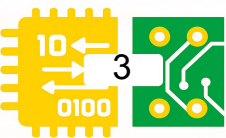
System Architecture

Six VHDL units: shared package, three debouncer instances, timer, FSM, display decoder, and structural top level. Button edges and timer signals are single-cycle pulses, keeping the control handshake free of level-sensitivity bugs.



Signal Flow

- Call buttons → debounce → FSM
- FSM ↔ timer via start/done handshake
- FSM → LEDs (up / down / door)
- FSM → 7-seg decoder → display AN0



Module Design

1

elevator_pkg

Shared declarations: 2-bit floor codes (U="00", E="01", 1="10"), state type (IDLE, MOVING, DOOR_OPEN), timer-mode type. Centralized encodings.

2

debouncer

Three stages: synchronizer, 10 ms debounce counter, rising-edge detector. One clean pulse per press. Instantiated once per button.

3

timer_unit

Parameterized down-counter: travel 2 s, door 3 s. Start pulse latches count; expiry yields single-cycle done pulse. Generics enable N-test technique.

4

seg7_decoder

Combinational lookup mapping floor code to active-low segment pattern (u, E, or 1). Single static digit AN0 — no refresh logic needed.

5

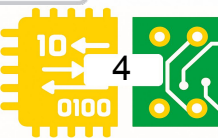
elevator_fsm

Three-state Mealy FSM with request latching, nearest-first scheduler, floor register, and motion control via timer handshake. Outputs direction LEDs and floor code.

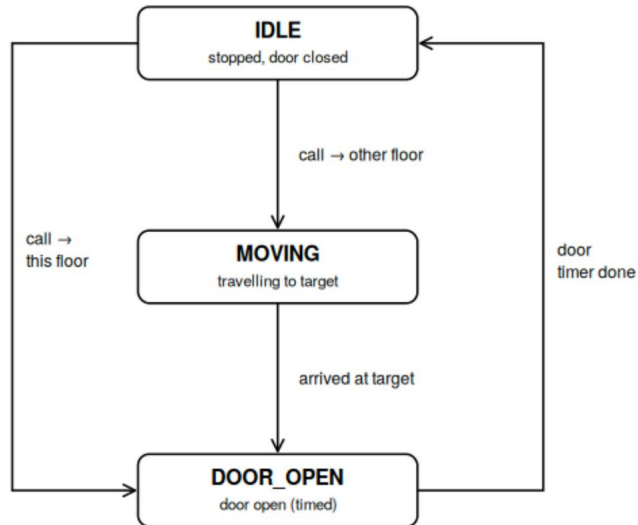
6

elevator_top

Structural integration of all modules. Inverts CPU_RESETN once. Hardware-scale generics exposed for testbench scaling.



FSM & Top-Level Integration

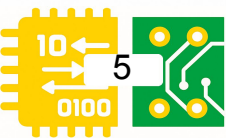


elevator_fsm – Control Core

- **Request latching:** captures presses every cycle, even mid-travel
- **Scheduler (IDLE):** nearest-first, tie-goes-up, one target per cycle
- **Mealy outputs:** direction LEDs assert one cycle before state update
- **Motion:** each travel-timer done steps floor register; on arrival, latch clears, enters DOOR_OPEN
- **Reset:** synchronous; clears latches, state IDLE, floor E

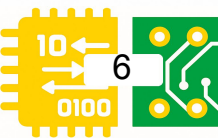
elevator_top

Structural integration of all modules. Inverts CPU_RESETN once. Hardware-scale generics exposed for testbench scaling.



Key Design Decisions

Decision	Choice	Rationale
FSM type	Mealy	Fewer states; reacts to call in same cycle
State count	3 (IDLE, MOVING, DOOR_OPEN)	Direction is output, not state — minimal machine
Scheduler	Nearest-first, tie-goes-up	Simple, deterministic, starvation-free
Reset	Single synchronous	All-synchronous design; no async recovery
Floor init	E (ground)	Natural demo start; board boots at E
Request storage	Latches inside FSM	Removes module boundary and cross-module bugs
Display	Single static digit (AN0)	Only one character shown — no refresh needed
Timer scaling	Generics (N-test)	One source for fast sim and full-scale hardware
RTL portability	No vendor primitives/IP	Synthesizes unchanged in Vivado and Lattice Diamond



Verification Results

Self-checking testbenches use VHDL `assert` with `severity error`; PASS/FAIL printed to console. N-test technique for fast runs. System testbench drives only raw button/reset ports, checking solely through output ports — exactly what a board observer sees.

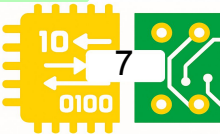
Module Testbenches

Testbench	Result
tb_timer_unit	PASS
tb_debouncer	PASS
tb_seg7_decoder	PASS
tb_elevator_fsm	PASS
tb_elevator_top	PASS

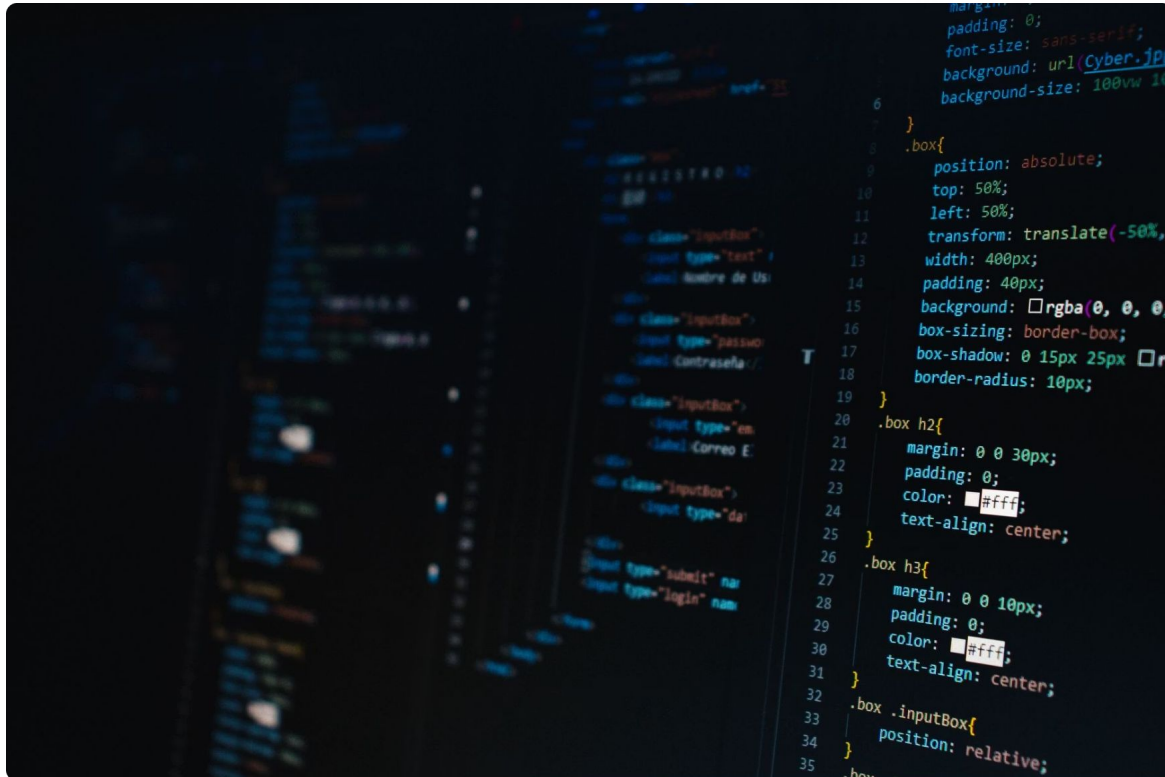
7 System Scenarios

#	Scenario
1	Reset → IDLE, display E
2	At E, call 1 → up, travel, door at 1
3	Multi-floor descent → down, E→U
4	Call current floor → door, no move
5	Bouncy press → one trip
6	Tie at E (U+1) → serve 1, then U
7	Full traverse U→E→1

✓ All 7 scenarios pass in simulation and on hardware.



FPGA Implementation – Nexys A7-100T



Resource Utilization (XC7A100T)

Negligible fabric usage justifies selecting a smaller FPGA for Phase 2 PCB.

Timing

- WNS: **+5.135 ns** vs. 10 ns constraint
- 0 failing endpoints — all constraints met

Demo

Board boots at E, all LEDs off. Seven scenarios exercised live, matching simulation.

71 LUTs

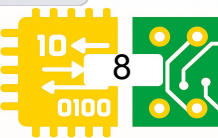
~1% of device

126 Regs

~1% of device

23 I/O

Bonded ports



Phase 2 PCB & Project Management

Custom PCB – Lattice MachXO2

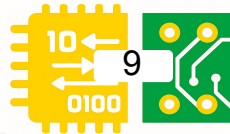
Standalone board in **Altium Designer** around LCMXO2-640HC-4TG100C (640 LUTs, 100-pin TQFP).

- Single 3.3 V supply — on-chip core regulation
- Instant-on, internal config flash — no external SPI memory
- Unchanged RTL — vendor-neutral VHDL re-synthesizes in Lattice Diamond

Remaining work: schematic, ERC, layout, DRC, BOM, Gerbers → /pcb.



Timeline





Team & Conclusion

Nnaemeka

VHDL/FSM lead — RTL, scheduler, integration, bring-up

Sumon

Verification — testbenches, simulation, sign-off

Suchi

Hardware demo — on-board testing, video

Oluwasholape

PCB lead — Altium schematic, layout, Gerbers

The 3-floor controller meets all MVP requirements — verified in simulation and hardware across seven scenarios. The three-state Mealy architecture kept the design minimal; generic-based timing enabled fast verification with full-scale behavior from one source; vendor-neutral RTL transfers directly to the MachXO2 PCB. Phase 2 hardware and final documentation remain.

